바이오메디컬 시스템반도체 융합설계 인력육성 센터

2025 Chip Design Contest

Pattern Generator for High-Speed Memory Test

Graduate School of Electronic and Electrical Engineering,
Kyungpook National University

Insu Jeong, Joowan Kang, Junghwan Kim, Sunghun Jung, and Byungin Moon

Introduction

The recent rapid growth in memory demand and advances in memory technologies have driven extensive research efforts focused on high-speed testing of high-capacity memories. The March algorithm is widely used for memory testing due to its simplicity and high fault coverage. In this study, we developed an ISA-based March pattern generator[1] implemented on an FPGA that generates test patterns according to stored March algorithm instructions, enabling flexible generation of various March patterns.

March Algorithm

The March algorithm diagnoses various kinds of defects by sequentially increasing or decreasing addresses and applying diverse test patterns for memory testing. By storing a specific March pattern value in each memory address and then reading it back to confirm consistency, the algorithm verifies the normal operation of each cell. Despite the simplicity of its operation combinations, it offers a high fault-detection rate and is widely employed for testing a variety of memories, including DRAM, SRAM, and flash memory.

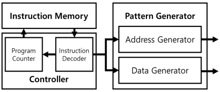

Fig. 1. Hardware architecture of the pattern generator

Hardware Architecture of Pattern Generator

The hardware architecture of the ISA-based March pattern generator consists of Instruction Memory, Controller, and Pattern Generator as illustrated in Fig. 1. The Instruction Memory stores instructions that constitute March algorithms. The Controller comprises a Program Counter and an Instruction Decoder, with the Instruction Decoder interpreting fetched instructions and providing control signals to both the Program Counter and Pattern Generator. The Program Counter generates the next instruction address based on these control signals. The Pattern Generator consists of an Address Generator and a Data Generator, which respectively output sequential memory addresses and corresponding data for read/write operations during testing.

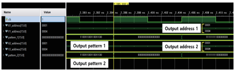

Fig. 2. Simulation waveforms of the pattern generator

Test Environment for FPGA Operation Verification

We designed an ISA-based March pattern generator using Verilog HDL and verified its functionality through simulation. Fig. 2 presents the simulation waveforms demonstrating linearly increasing memory addresses and correct test pattern generation. The proposed design was implemented on a Xilinx xc7z020clg484-1 FPGA and validated through UART serial communication. Fig. 3 shows the UART serial communication results, and Table 1 reports the hardware resource utilization. Based on the verified design, the final chip layout is illustrated in Fig. 4.

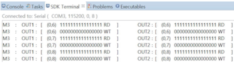

Fig. 3. UART serial communication results of the pattern generator

Resource	Utilization
LUT	2364
LUTRAM	60
FF	5665
BUFG	2

Table 1. Resource utilization of the pattern generator

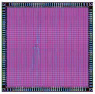

Fig. 4. Layout of the pattern generator ASIC

Conclusion

This work presents an FPGA-based pattern generator for high-speed memory test. The FPGA implementation consumes minimal hardware resources, enabling significant reductions in both cost and power for test equipment. An ASIC with the same architecture is now being fabricated. After the chip is completed, its functionality and performance will be verified using the evaluation setup shown in Fig. 5.

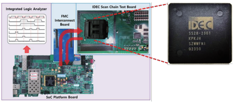

Fig. 5. Pattern generator ASIC verification environment

Reference

[1] I. Jeong, J. Kang, T. Kim, S. Jo, and B. Moon, “FPGA Implementation and Verification of an ISA-Based March Pattern Generator,” in Proceedings of KIIT Conference, 2024, pp. 204-204.

Acknowledgment

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